

FDSN existing stations - SoH

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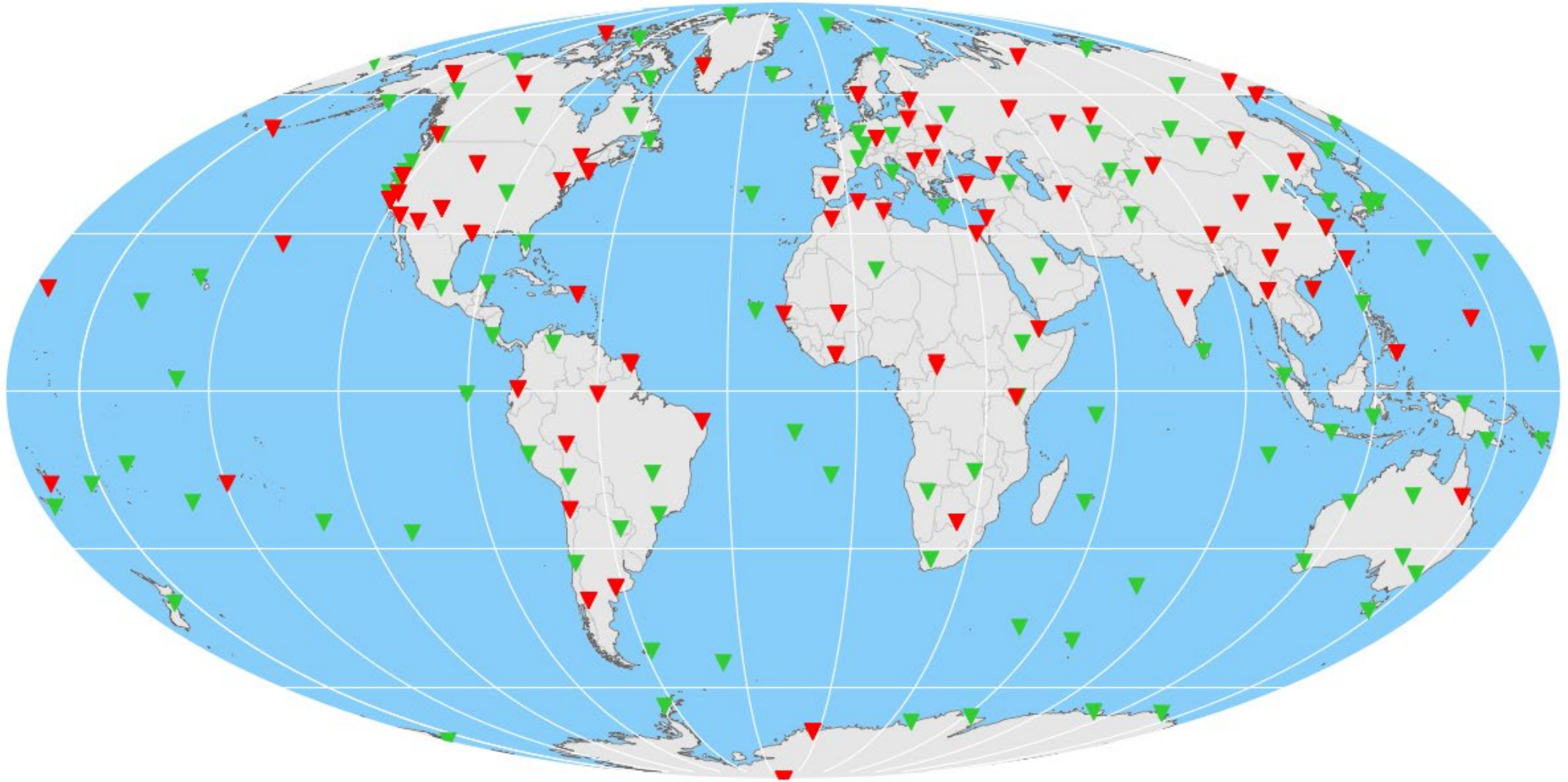
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_FDSN – virtual network, IRIS

~ 200 stations

Already closed stations

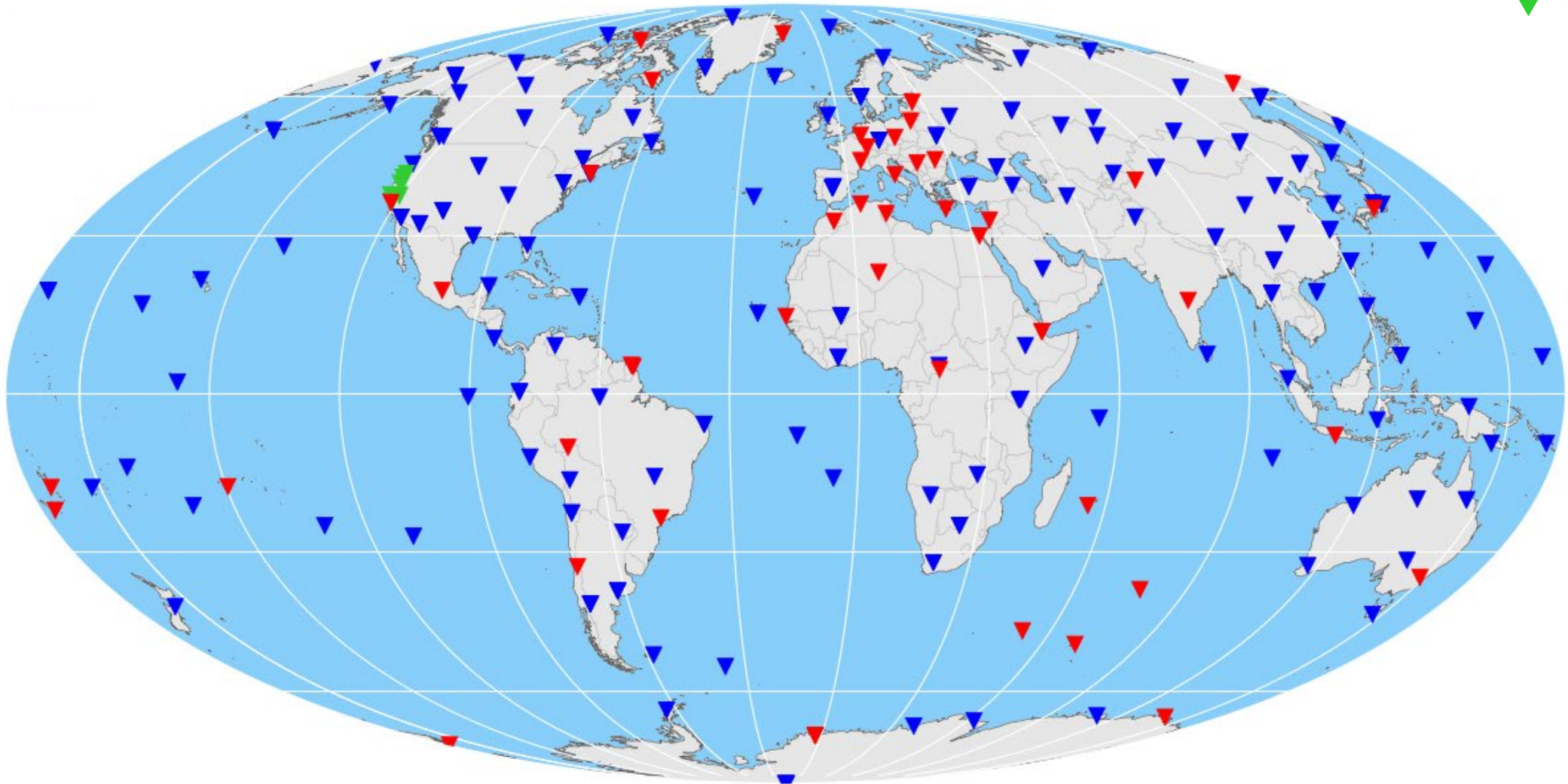
- ▼ already closed
- ▼ existing up to now



IDEA: find new stations with as close locations to the old ones as possible

Archives

- ▼ EIDA
- ▼ IRIS
- ▼ NCEDC



IDEA: find new stations with as close locations to the old ones as possible

Station SoH

- using quality metrics already computed in archives:

EIDA – wfcatalog

IRIS – mustang

NCEDC – none

(but their secondary archive is IRIS)

Daily means – fast and powerful tool for QC

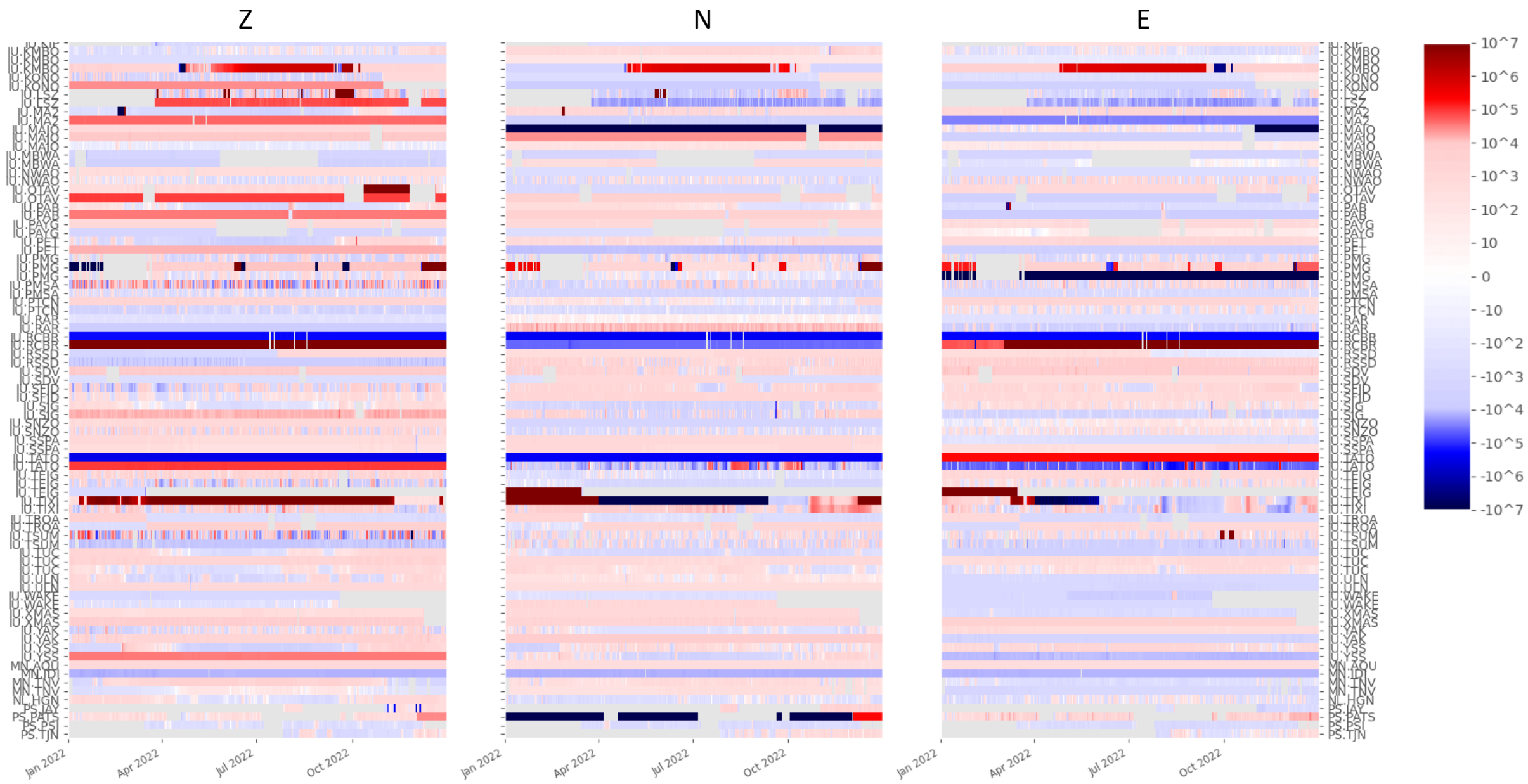
Probabilistic power spectral density (PPSD)

- time-frequency representation of the signal
- not included in EIDA WFCatalog (yet)
- need to download continuous data
- long computation of PPSD
- depends on metadata (changes/updates!)

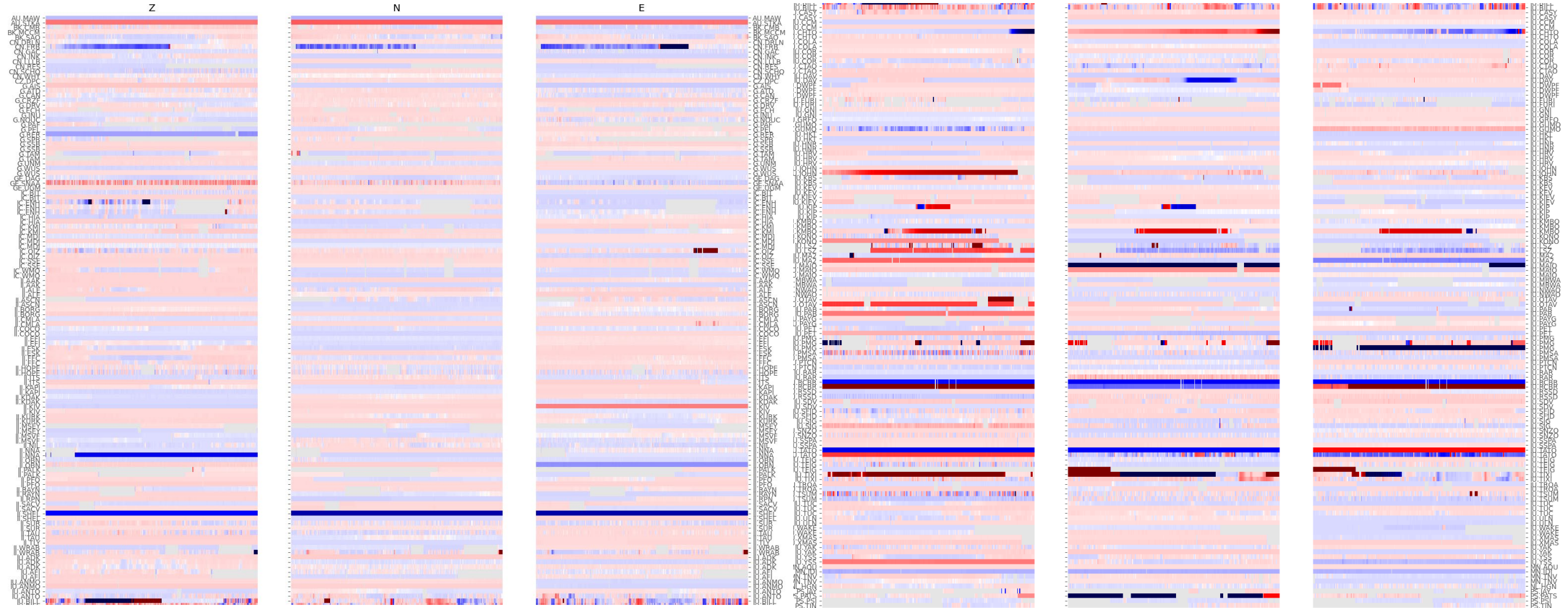
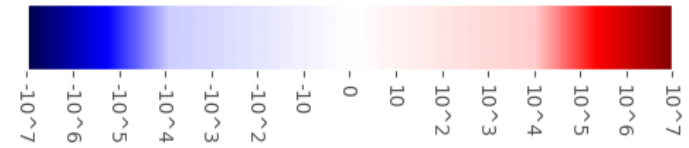
Metrics in WFCatalog

- no frequency analysis
- easy and fast to download from EIDA
- doesn't depend on metadata
- metrics:
 - daily median
 - daily std
 - daily max, min
 - ...

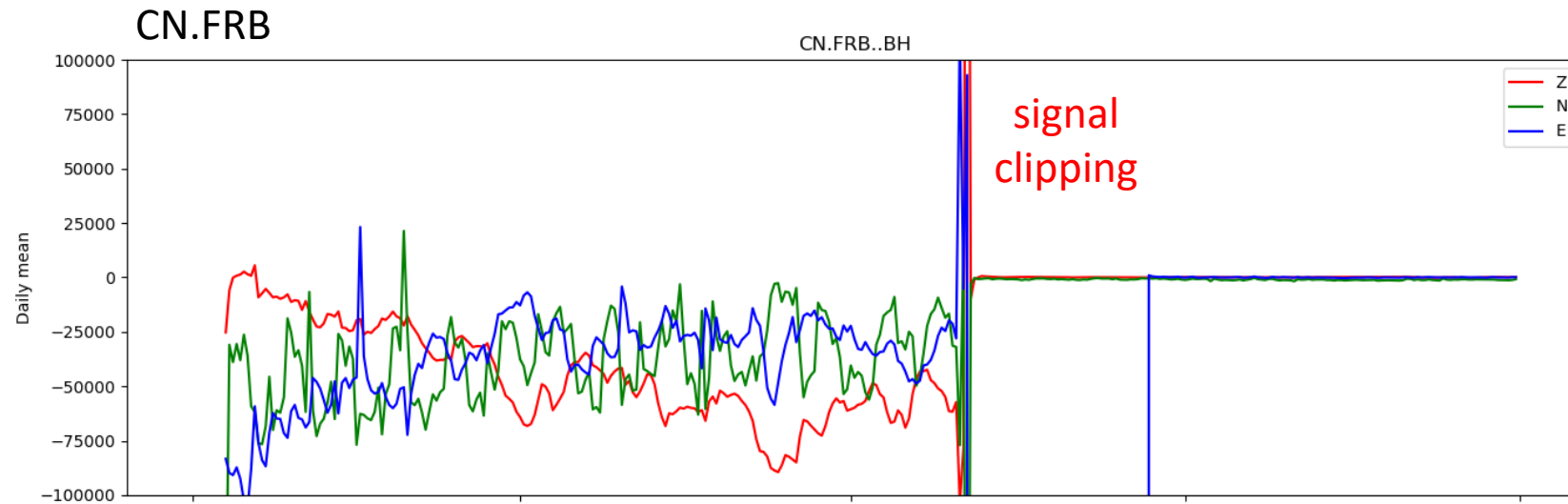
Saturation of signal daily medians/offsets



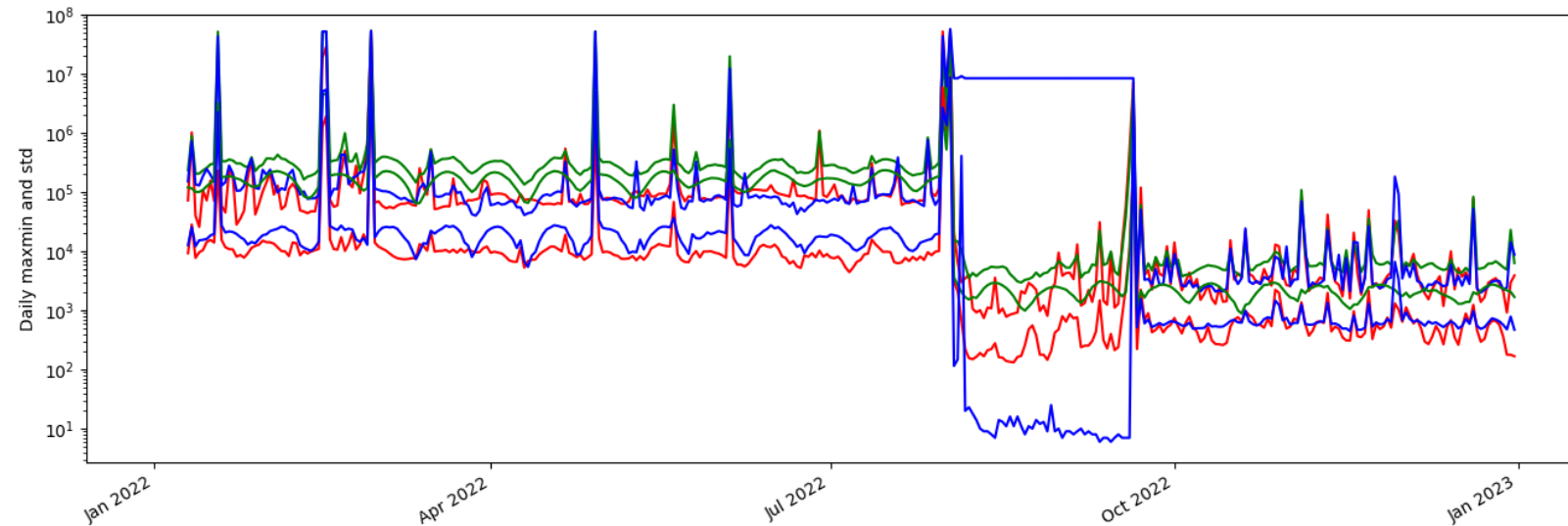
_FDSN - daily medians



Inspection of individual station



MEDIANS

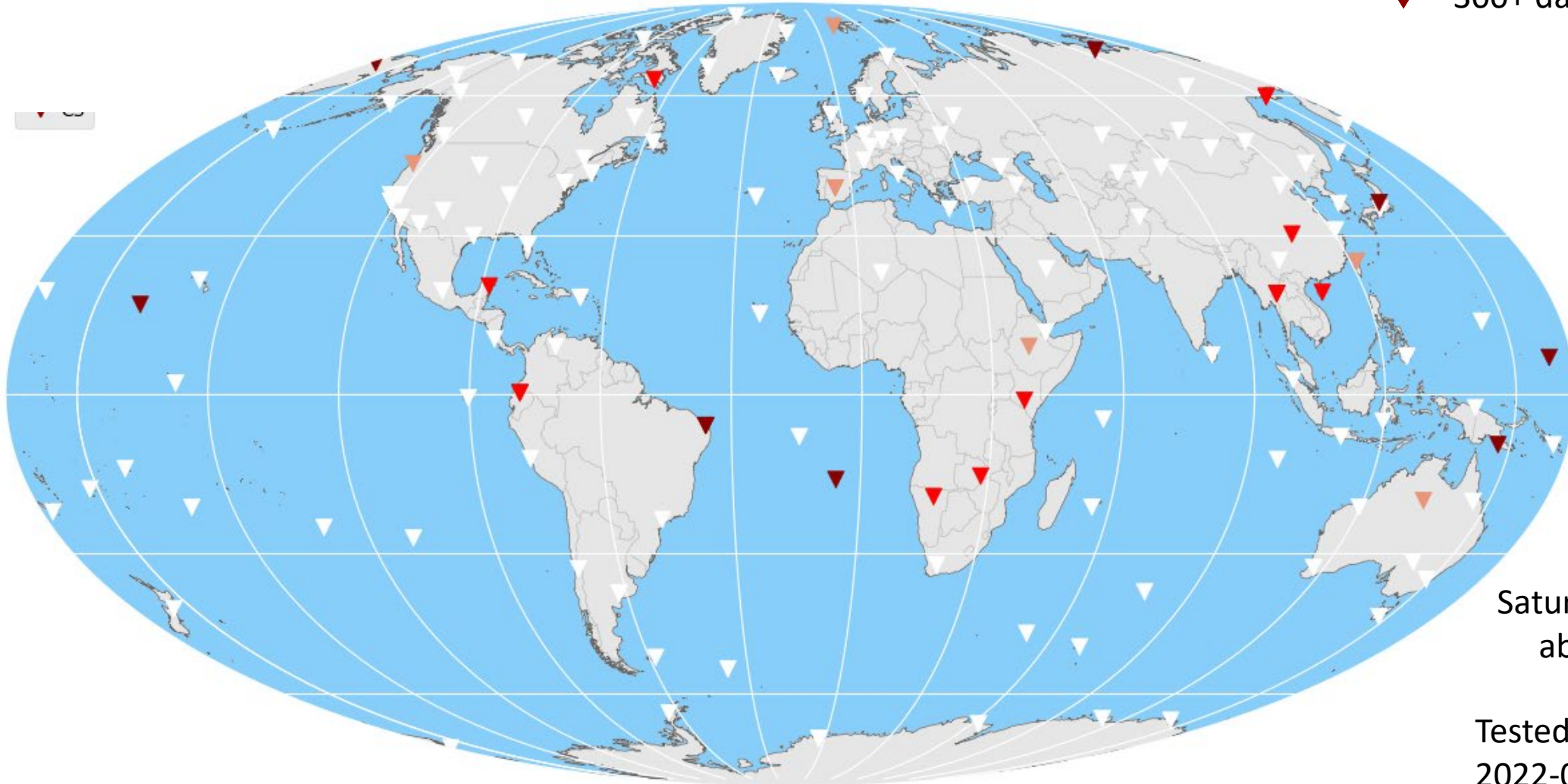


MAXMINs = MAX - MIN

STDs

Saturated medians - map

- ▼ all stations (137)
- ▼ 10+ days: 24/137 18 %
- ▼ 100+ days: 18/137 13 %
- ▼ 300+ days: 8/137 6 %



Saturation condition:
 $\text{abs}(\text{median}) > 1\text{e}6$

Tested period:
2022-01-01 – 2023-01-01

IDEAS

- preparing SoH report periodically (each year?) (and contacting operators?)
- quality metrics:
 - median (maxmin, std)
 - PPSD (not available for all stations)
 - data availability
- metadata checks
- sensor orientation
- ...